



**Software Engineering Department
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User Guide

Characterizing Normal Brain Development via EEG Pattern Matching

CODE: 26-1-R-11

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<https://github.com/wadiea1/Eeg-age-patterns.git>**

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Guide focus: This User Guide explains how to operate the notebook in the nominal successful flow. It intentionally avoids detailed debugging and internal implementation maintenance, which belong in the Maintenance Guide.

1. Purpose and Scope

This User Guide explains how to run the EEG Motif Discovery Pipeline developed for the project Characterizing Normal Brain Development via EEG Pattern Matching. The system is implemented as a Google Colab notebook and is intended to help the user move from HBN-EEG recordings to interpretable motif results.

The pipeline supports the main research goal of comparing normal EEG development across ages 5-21. Instead of comparing raw EEG amplitudes directly, the system focuses on short repeated signal shapes, called motifs, and compares their occurrence, diversity, and average shape across age groups [1], [6], [9].

The guide is written for users who already have access to the project Google Drive folder, the HBN-EEG dataset, and the final notebook. It describes the normal successful operating path: configure, run, generate outputs, and understand the plots.

2. System Overview

The EEG Motif Discovery Pipeline is an automated research workflow. It loads EEG recordings, preprocesses the signal, segments selected channels into short windows, converts each window into shape-based codes, identifies recurring motifs, and generates result plots.

Stage	User action	Main output
Stage 1 - Preprocessing	Run the preprocessing cell after setting paths, tasks, and channels.	Cleaned, normalized EEG signals saved in 01_cleaned/.
Stage 2 - Windowing and Encoding	Choose one or more L values and run the encoding cell.	Window arrays, CTM codes, OPM codes, and channel tracking saved in 02_codes_L{L}/.
Stage 3 - Motif Libraries	Build motif libraries using the selected IN/OUT rule.	Per-group and per-channel motif libraries saved in 03_libraries_L{L}/.
Stage 4 - Analysis and Plots	Run statistical tests, diversity plots, motif-shape plots, and motif-frequency plots.	Figures and summaries saved in 04_results_L{L}/.

Two complementary encodings are used. OPM captures the relative rank order of values inside a window, so it focuses on local shape rather than amplitude. CTM captures the global structure of peaks and valleys using Cartesian-tree structure. Together, they allow the same EEG segment to be analyzed from local-order and global-structure perspectives [5], [6], [8].

3. Requirements

- A Google account with access to the shared project folder in Google Drive.
- Google Colab runtime; High-RAM runtime is recommended for long preprocessing runs.
- HBN-EEG dataset folder ds005505 stored in Google Drive in BIDS-style structure.
- The final notebook: EEG_Pipeline_final_project_TASKs.ipynb.
- Python packages installed by the notebook: mne, mne-icalabel, numpy, pandas, scipy, matplotlib.
- Enough Drive space for cached outputs: cleaned signals, encoded windows/codes, motif libraries, and generated plots.

Important: Stage 1 is the slowest part because it includes filtering and ICA/ICLabel artifact handling. The pipeline uses Drive caching so later experiments can reuse cleaned data instead of repeating preprocessing.

4. Dataset and Input Structure

The input dataset is HBN-EEG, an analysis-ready EEG dataset in BIDS format. It contains high-density EEG recordings for participants aged 5-21 and includes metadata files such as participants.tsv, channels.tsv, events.tsv, and EEG sidecar files [1], [10].

Each participant folder contains EEG task files in EEGLAB .set format. The notebook uses participants.tsv to assign subjects to age groups and searches the dataset folders for the selected task recordings.

Item	Expected location / description
Dataset root	base_path should point to the ds005505 folder in Google Drive.
Participant metadata	participants.tsv at the dataset root. Used to read ages and split subjects into groups.
EEG recording	sub-<id>/eeg/sub-<id>_task-<TaskName>_eeg.set
Optional metadata	events.tsv, channels.tsv, and eeg.json can exist next to the .set file.
Output root	WORK_ROOT points to the project output folder, such as /content/drive/MyDrive/eeg_pipeline_outputs.

The analysis can be applied to passive tasks such as RestingState and movie watching (for example, DespicableMe), and can also be extended to active tasks such as contrastChangeDetection by adding the correct task name to the TASKS list.

5. Current Notebook Configuration

Most user-controlled settings are located in Cell 1. Before running the pipeline, the user should verify the paths, task list, channel list, window-length grid, and motif rule.

Parameter	Meaning for the user	Current / typical value
base_path	Path to the HBN-EEG ds005505 dataset folder in Drive.	Project Drive path ending with ds005505.
WORK_ROOT	Folder where all pipeline outputs are saved.	/content/drive/MyDrive/eeg_pipeline_outputs
TASKS	Tasks to process. Add or remove task names here.	Default notebook value may be ["RestingState"]. Results can also be generated for DespicableMe and contrastChangeDetection when included.
RUN_MODE	FULL uses all available subjects; SCAN uses a smaller sample per group.	FULL
CHANNELS_45 / FRONTAL_CHANNELS	Selected representative channels used in the analysis. The variable name is kept for compatibility even if the list is expanded.	Representative E-number channels across the scalp.
FILTER_L / FILTER_H	Bandpass filtering range for preprocessing.	1.0 Hz to 45.0 Hz
ICA_N_COMPONENTS	Number of ICA components used for artifact removal.	15
OVERLAP	Overlap ratio between windows.	0.0 for non-overlapping windows
WINDOW_LENGTHS_GRID	List of L values to scan or encode.	[8, 10] in the current notebook; the project explored a wider range.
L_FOR_PLOTS	Selected L value for final plots.	10
IN_GROUP_PCT / OUT_GROUP_PCT	Motif presence rule for the target group and the other groups.	Set in Cell 1 and used when building libraries.
MAX_SECONDS_PER_SUBJECT	Drive-safe duration limit during processing.	120 seconds

Parameter exploration: During development, several settings were tested, including different age-group splits, IN/OUT rules, and window lengths. The reported plots mainly use L = 10 because it gave clear motif outputs while still preserving short-window structure.

6. Quick Start - Recommended Successful Run

1. Open the final notebook in Google Colab.
2. Connect to a runtime. Use High-RAM if you plan to preprocess many subjects.
3. Run Cell 1 to install packages, mount Google Drive, define paths, and load the main parameters.
4. Check that `base_path` points to `ds005505` and that `WORK_ROOT` points to the desired output folder.
5. Run Cell 3 to load `participants.tsv` and split participants into age groups.
6. Run Cells 4 and 5 to define and test CTM and OPM encoders.
7. Run Cell 6 to preprocess signals and save cleaned files to `01_cleaned/`.
8. Run Cell 7 for the desired window length values in `WINDOW_LENGTHS_GRID`.
9. Run Cell 8 to build motif libraries for the chosen `L`.
10. Run Cell 8.5 if you want to compare several `L` values and select the best one.
11. Run Cell 9 for per-channel statistical tests.
12. Run Cell 10 and Cell 11 to generate diversity and motif-shape plots.
13. Run Cell 12 to generate motif-frequency plots per channel.
14. Open `04_results_L{L}/` in Drive and review the saved plots.

Recommended first run: Start with one task and a short runtime setting to confirm that the dataset path, participant split, and output folders are correct. After the pipeline produces valid outputs, expand to the full run.

7. Notebook Cells - User Operating Guide

The notebook is organized into named cells. The user should run them in order for a complete successful workflow. The table below describes what each cell does from the user perspective.

Notebook cell	What the user gets	When to run
Cell 1 - Setup	Installs packages, mounts Drive, sets dataset/output paths, defines tasks, selected channels, preprocessing parameters, IN/OUT rule, overlap, and <code>L</code> grid.	Run first every time. Re-run after changing paths or parameters.
Cell 2 - Optional cleanup	Deletes Stage 2 and Stage 3 cached outputs while keeping Stage 1 cleaned files.	Use only when rebuilding codes or motif libraries.
Cell 3 - Age groups	Loads <code>participants.tsv</code> , keeps ages 5-21, and splits participants into balanced quantile age groups.	Run before preprocessing and all later stages.
Cell 4 - CTM encoder + test	Defines the CTM parent-distance representation and validates it with a small unit test.	Run once before encoding windows.
Cell 5 - OPM encoder + test	Defines the OPM natural rank-order representation and validates it with a small unit test.	Run once before encoding windows.
Cell 6 - Stage 1 preprocessing	Loads each <code>.set</code> file, applies filtering, average reference, ICA + ICLabel, channel selection, z-score normalization, and saves cleaned data.	Run when preprocessing is needed or when channels/filter settings changed.
Cell 7 - Stage 2 encoding	Slices cleaned signals into fixed-length windows for each <code>L</code> , computes CTM/OPM codes, tracks channel indices, and saves <code>.npz</code> files.	Run after preprocessing or after changing <code>L/overlap/duration</code> .
Cell 8 - Stage 3 motif libraries	Builds motif libraries per task, age group, channel, and encoding type using the IN/OUT rule.	Run after Stage 2 and before generating motif plots.
Cell 8.5 - Grid search	Scans candidate window lengths and prints motif statistics so the user can choose a useful <code>L</code> .	Run during parameter selection.
Cell 9 - Statistical tests	Runs Kruskal-Wallis tests per channel for the selected <code>L</code> and saves/prints statistical summaries.	Run before final result review.
Cell 10 - Diversity and motif-shape plots	Generates per-channel diversity boxplots and motif-shape plots for CTM/OPM.	Run after motif libraries exist.

Cell 11 - Motif shape + consistency check	Creates additional motif mean/SEM plots and verifies that encoded windows have the expected L.	Run as a validation step for final plots.
Cell 12 - Motif frequency per channel	Computes the median percentage of motif windows per channel and group, then saves CTM/OPM frequency bar plots.	Run last for final poster/report plots.

8. Outputs Saved by the Pipeline

The notebook saves outputs by processing stage. This makes the workflow reproducible and avoids repeating expensive computation when only plots or motif rules change.

Folder	Content	How the user uses it
01_cleaned/	Cleaned EEG signals per subject and task after filtering, ICA/ICLabel, selected channel extraction, and z-score normalization.	Reuse these files when only changing L, motif thresholds, or plots.
02_codes_L{L}/	Window arrays, CTM codes, OPM codes, channel indices, and channel names for each subject/task at a specific L.	Used by motif library building, diversity calculations, and frequency analysis.
03_libraries_L{L}/	Pickle files containing motif sets for each task, group, channel, and encoding type.	Required before motif-shape and motif-frequency plots.
04_results_L{L}/	Saved figures, statistics, diversity plots, motif-shape plots, and motif-frequency plots.	Main folder to inspect for report/poster outputs.

9. Understanding the Result Plots

The result plots compare age groups for the same task, channel, and window length L. They are designed to be interpretable: the user can see whether a group has more varied codes, whether a motif shape differs between groups, and which channels contain motifs more often.

9.1 Diversity per subject

Diversity is computed for each subject and channel as: number of unique codes divided by total number of codes. A higher value means the subject has more different encoded window shapes; a lower value means the signal repeats fewer code types more often. Diversity is computed separately for CTM and OPM because the encodings describe shape differently.

9.2 Motif-shape plots

Motif-shape plots show the average raw EEG shape, usually mean plus/minus SEM, for windows that matched motif codes. If only one group appears, that group was the one that produced motifs passing the selected rule. If multiple groups appear, their average motif shapes can be compared across the window positions.

9.3 Motif-frequency plots

Motif-frequency plots show channels where motif codes were found. For each channel and group, the y-axis reports the median percentage of that group's windows that matched motifs. Higher bars indicate that motifs recur more often for the typical subject in that group and channel.

9.4 CTM versus OPM interpretation

Encoding	What it emphasizes	How to interpret results
OPM	Relative rank order of points inside the window; local order and contour.	Useful for detecting local shape differences even when amplitudes vary between subjects.
CTM	Global organization of peaks and valleys represented by a Cartesian-tree structure.	Usually more selective; when motifs appear, they indicate repeated global structural patterns.

Motifs and diversity should be read together. Motifs represent repeated codes, while diversity represents how many different codes appear. If diversity is high, the EEG windows are spread across more code types, which may reduce the number of repeated motifs. If diversity is low, repeated motifs may be more common.

10. Efficient Re-run Workflow

The user usually does not need to rerun the full notebook after every change. Use the table below to rerun only the required stages.

What changed?	Minimum cells to rerun	Reason
Only need to regenerate plots for the same L and same libraries.	Cells 10-12	Cleaned data, codes, and libraries already exist.
Changed L grid or selected L.	Cell 7 -> Cell 12	Windows and codes depend on L.
Changed overlap or maximum seconds per subject.	Cell 7 -> Cell 12	Window slicing changes, so codes and motif libraries must be rebuilt.
Changed IN/OUT rule or motif presence rule.	Cell 8 -> Cell 12	Motif libraries depend on the rule, but cleaned data and codes do not.
Changed selected channel list.	Cell 6 -> Cell 12	Stage 1 extracts and saves the selected channels.
Added a new task to TASKS.	Cell 6 -> Cell 12 for that task	The task must be preprocessed, encoded, and analyzed.
Changed filtering or ICA parameters.	Cell 6 -> Cell 12	Preprocessed signals must be rebuilt.

11. User Checklist Before Submission or Demo

- The notebook runs from Cell 1 to Cell 12 without stopping in the intended configuration.
- The selected TASKS list matches the tasks reported in the poster/report.
- The selected L value in plots matches the reported L value.
- 04_results_L{L}/ contains diversity plots, motif-shape plots, statistical summaries, and motif-frequency plots.
- Plot titles show the correct task, channel, encoding type, group rule, and L value.
- The reported age groups and selected channels match the configuration used when generating the final plots.
- GitHub contains the final notebook, the project documents, the poster, and any supporting output examples.

Demo tip: For a short demonstration, show Cell 1 parameters, the staged output folders, one diversity plot, one motif-shape plot, and one motif-frequency plot. Then explain how the same notebook can be rerun for another task or L value.

12. References

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